Smart Hospital Finder using AI Algorithms

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Smart Hospital Finder using AI Search Algorithms

• INTRODUCTION

This project is a Command Line Interface (CLI) based application designed to help users find the nearest or cheapest hospital. It uses Artificial Intelligence algorithms like Breadth First Search (BFS) and Uniform Cost Search (UCS) to provide efficient and optimized results. The system works on a graph-based structure where locations are connected, and hospitals are identified within the network.

• MOTIVATION OF THE PROJECT

In emergency situations, people often struggle to find the nearest or affordable hospital quickly. This project is motivated by the need to provide a fast, simple, and reliable solution for hospital search using AI techniques.

• PROBLEM STATEMENT

Users face difficulty in identifying the nearest or most cost-effective hospital within limited time, especially during emergencies.

• OBJECTIVE OF THE PROJECT

- To find the nearest hospital using BFS
- To find the cheapest hospital using UCS
- To provide a simple CLI-based interface
- To simulate real-world hospital search scenarios

• EXISTING METHODS

- Google Maps
- Manual searching
- Online directories

• PROS AND CONS OF THE STATED METHODS

Pros:

- Easy to access
- Provides real-time data

Cons:

- Requires internet
- Not customizable
- Can be time-consuming

• HARDWARE AND SOFTWARE REQUIREMENTS

Hardware:

- Laptop / Computer

Software:

- Python
- VS Code / Terminal
- GitHub

• METHODOLOGY AND GOAL

The system represents locations as a graph. Each node represents a location, and edges represent connections between them.

- BFS is used to find the nearest hospital
- UCS is used to find the lowest cost hospital

The goal is to provide efficient and accurate results.

• FUNCTIONAL MODULES DESIGN AND ANALYSIS

- User Input Module
- BFS Search Module
- UCS Search Module

- Data Handling Module
- Output Display Module

• ALGORITHM DEVELOPMENT

Breadth First Search (BFS):

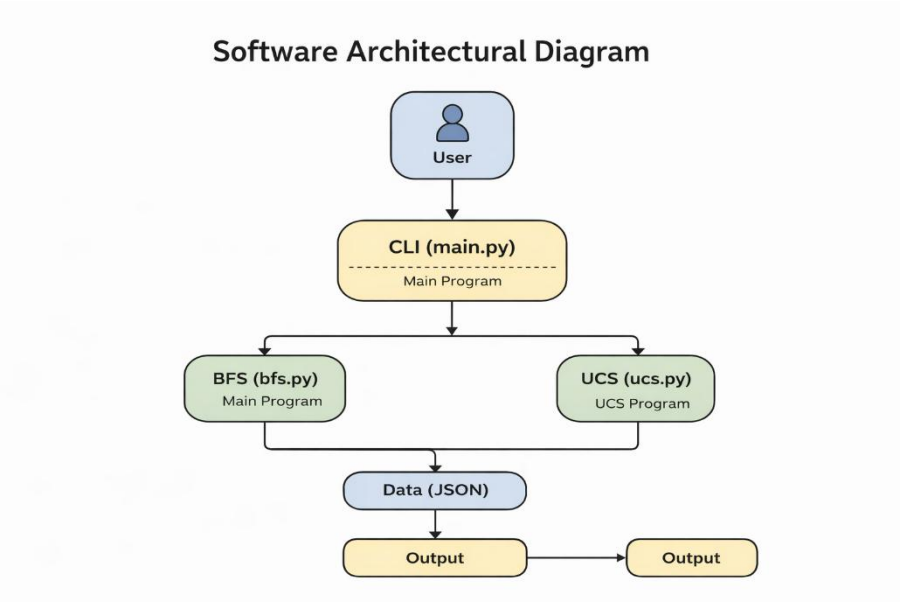
Traverses the graph level by level to find the nearest hospital.

Uniform Cost Search (UCS):

Finds the hospital with the minimum cost using weighted paths.

• SOFTWARE ARCHITECTURAL DIAGRAM

User → CLI (main.py) → BFS/UCS → Data (JSON) → Output



• CODING

The project is implemented in Python using multiple files:

- main.py (main program)
- bfs.py (BFS logic)
- ucs.py (UCS logic)
- utils.py (helper functions)

- data.json (data storage)

• OUTPUT

```
PS C:\Users\Gauri Chaudhary\Videos\Smart Hospital Finder> & "C:
t Hospital Finder/main.py"

===== Smart Hospital Finder 🚀 =====
1. Find Nearest Hospital (BFS)
2. Find Cheapest Hospital (UCS)
3. View Hospitals
4. Emergency Mode 🚑
5. Exit
Enter choice: 1

Available Locations:
1. MP Nagar
2. Habibganj
3. Arera Colony
4. Kolar Road
5. BHEL
6. Govindpura
7. AIIMS Bhopal
8. Hamidia Hospital
Select location (number): 1

Select Hospital Type:
1. Government
2. Private
3. All
Enter choice: 1

✅ Nearest Hospital: AIIMS Bhopal at AIIMS Bhopal
Path: MP Nagar → Arera Colony → Kolar Road → AIIMS Bhopal
```

```
===== Smart Hospital Finder 🚀 =====
1. Find Nearest Hospital (BFS)
2. Find Cheapest Hospital (UCS)
3. View Hospitals
4. Emergency Mode 🚑
5. Exit
Enter choice: 2

Available Locations:
1. MP Nagar
2. Habibganj
3. Arera Colony
4. Kolar Road
5. BHEL
6. Govindpura
7. AIIMS Bhopal
8. Hamidia Hospital
Select location (number): 1

Select Hospital Type:
1. Government
2. Private
3. All
Enter choice: 2

💰 Cheapest Hospital: Noble Hospital at Arera Colony
Total Cost: 603
Path: MP Nagar → Arera Colony
```

```
===== Smart Hospital Finder 🚀 =====
1. Find Nearest Hospital (BFS)
2. Find Cheapest Hospital (UCS)
3. View Hospitals
4. Emergency Mode 🚑
5. Exit
Enter choice: 3

Available Hospitals:
AIIMS Bhopal | Location: AIIMS Bhopal | Cost: 200 | Type: Government
Hamidia Hospital | Location: Hamidia Hospital | Cost: 100 | Type: Government
Noble Hospital | Location: Arera Colony | Cost: 600 | Type: Private
Bansal Hospital | Location: Habibganj | Cost: 800 | Type: Private
```

```
===== Smart Hospital Finder 🚀 =====
1. Find Nearest Hospital (BFS)
2. Find Cheapest Hospital (UCS)
3. View Hospitals
4. Emergency Mode 🚑
5. Exit
Enter choice: 4

Available Locations:
1. MP Nagar
2. Habibganj
3. Arera Colony
4. Kolar Road
5. BHEL
6. Govindpura
7. AIIMS Bhopal
8. Hamidia Hospital
Select location (number): 1

Select Hospital Type:
1. Government
2. Private
3. All
Enter choice: 1

🚑 EMERGENCY! Nearest Hospital: Bansal Hospital at Habibganj
Path: MP Nagar → Habibganj
```

• KEY IMPLEMENTATIONS OUTLINE OF THE SYSTEM

- CLI-based interaction
- Graph-based structure
- AI search algorithms
- Input validation

• SIGNIFICANT PROJECT OUTCOMES

- Efficient hospital search
- Practical application of AI algorithms
- Improved decision-making in emergencies

• TESTING AND REFINEMENT

The system was tested with different inputs and locations. Invalid inputs were handled properly to ensure smooth execution.

• PROJECT APPLICABILITY ON REAL-WORLD APPLICATIONS

- Emergency response systems
- Healthcare navigation
- Smart city solutions

• CONTRIBUTION / FINDINGS OF THE PROJECT

The project demonstrates how AI algorithms can be used to solve real-world problems effectively. It shows the importance of graph traversal techniques in decision-making systems.

• LIMITATION / CONSTRAINTS OF THE SYSTEM

- Uses static data
- Limited number of locations
- No real-time updates

• CONCLUSIONS

The project successfully implements AI algorithms to solve the problem of hospital search. It provides a simple and efficient CLI-based solution.

• FUTURE ENHANCEMENTS

- Add real-time hospital data
- GPS integration
- GUI-based system
- Mobile application

• REFERENCES

- Python Documentation
- Course Materials (AI & ML)
- Online resources